

Executive Summary

Compliance Score

59 / 100HIGH RISK

Findings breakdown: 1 HIGH, 2 PASS.
Active threats: Group 2 (1024-bit) provides <112 bits of security (Logjam).

Key Findings

[HIGH] Group 2 (1024-bit) provides <112 bits of security (Logjam).

Replay Protection Readiness

ESP Sequence Number Check: MONOTONIC
ESP sequence numbers are monotonically incrementing.

Traffic Analysis Resistance

Metadata Exposure: HIGH (92.5% classifier confidence)
Tunnel is highly transparent to traffic analysis.

Key Lifetime

Not assessable: SA lifetimes are local policy per RFC 7296 section 2.8.

Caveats & Limitations

* PFS status could not be determined: no CREATE_CHILD_SA rekey observed.

Traffic Classification (ML)

Predicted Traffic Type: VOIP
Classification Confidence: 92.5%
A passive adversary can classify this tunnel's encrypted traffic as 'voip' with 92.5% confidence using flow-level metadata alone. This represents significant metadata leakage.

Countermeasure Simulation

Simulated Traffic Flow Confidentiality (TFC) padding countermeasures:
MTU Padding: 92.5% -> 77.0% confidence | 14.2% bandwidth overhead
Adaptive Padding: 92.5% -> 87.3% confidence | 17.0% bandwidth overhead

Recommended Actions

[HIGH] Upgrade DH group to Group 19 (ECP-256) or Group 20 (ECP-384). Groups 1/2 provide less than 112 bits of security and are vulnerable to the Logjam attack.

Executive Security Assessment

[MEDIUM] Deploy traffic padding countermeasures (RFC 4303 TFC padding, MTU padding, or adaptive padding) to reduce metadata leakage.
Current classifier confidence: 92.5%.